

Dmitri Volkov

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EDUCATION

Indiana University, Bloomington IN, US

2025 - 2026

M.S. in Computer Science

Research thesis in programming language theory (PLT): “Polymorphic Bottom-Up Weighted Relational Programming” with correctness proofs. Courses in PLT, modern algebra, FPGAs. GPA: 3.7.

Indiana University, Bloomington IN, USA

2021 - 2025

B.S. in Computer Science and B.M. in Music Composition

Highest distinction and honors. Minors in Computer Music and Mathematics, Computer Science concentrations in Programming Languages and Foundations. GPA: 3.99. Credits taken: 219.

PROJECTS

semiringKanren, logic/relational programming language

2025 - present

Typed polymorphic logic language computing over semirings. Can outperform faster-miniKanren, supports disequality constraints. Presented at ICFP 2025. github.com/sporkl/semiringkanren

Pivotuner, commercial audio plugin for macOS and Windows

2021 - present

Virtual instrument pure intonation autotune in C++. Developed with Grammy-winning/nominated musicians e.g. Jacob Collier, Nick Revel. \$2500+ in sales. Presented at MOXsonic 2024, SPLICE VI.

Hardware-accelerated matrix-vector multiplier (course project)

2025

32-bit float matrix-vector multiplier for Pynq-Z1, implemented in Verilog. Uses pipelined float FMAC IPs, supports arbitrary parallelization (up to hardware) through parametrization, beats NumPy.

Formally-verified finger tree (course project)

2025

Finger tree data structure verified in Rocq, including property-based tests and proofs of: operations preserving invariants, algebraic properties, and abstraction to lists.

Lisp-like language compiler (course project)

2024

Compiler from Lisp-like language to x86-64 assembly. Features: garbage collection, tail call optimization, compile-time evaluation, support for multithreaded programs.

Malt Transformer, transformer neural network implementation

2023

GPT architecture implementation as used by LLM AI models. github.com/sporkl/malt-transformer

SKILLS

Programming Languages: C/C++, OCaml, Java, Kotlin, TypeScript/JavaScript, (System)Verilog, Rocq, SQL, Python, Lua, R, PHP, Racket, Mathematica, Max/MSP/Pure Data, LaTeX, HTML/CSS, C#, x86.

Frameworks/Libraries: PostgreSQL, JUCE, UNIX libc, igrph, Arduino, React Native/Expo, LOVE2d, Android(X), Vivado, Qiskit, NumPy, Pandas, scikit-learn, websockets, Azure Functions, Snowflake.

Specializations: formal methods/verification, realtime digital signals/audio/music, compilers, math, functional and logic/relational programming, embedded/FPGAs/RTL design/HDL, quantum.

Miscellaneous: UNIX shell, debugging, optimizing, technical writing, research, devops, Agile, Vim, VSCode, JetBrains IDEs, Xcode, macOS, Linux, Windows, CMake, Git, web hosting, Microsoft Azure, APIs, Spanish, music (composition/performance/production).

EXPERIENCE

Internship at Old Road Software, Ridgefield CT, USA 2025
Full-stack development across iOS/Android/web/backend APIs, including OpenAI API integration. Projects: legaltech form generator, nutrition supplement counterfeit scanner, AI wellness app.

Undergraduate Instructor at Indiana University, Bloomington IN, USA 2023 - 2026
Courses: CSCI-C311/H11/A595 Programming Languages, CSCI-A290/A590 Straight Line to Deep Learning. Review code, solidify student understanding, collaborate to develop course materials.

PUBLICATIONS / PRESENTATIONS

Committing to the bit: Relational programming with semiring arrays and SAT solving 2025
Introduces semiringKanren logic language including denotational semantics, correctness-preserving SAT compilation, and applications. Presented at ICFP 2025. doi.org/10.48550/arXiv.2509.22614

Sonifications of Quantum Superpositions: Methods and Musical Applications 2025
First chapter of Advances in Quantum Computer Music, published by World Scientific. Illustrates general approach to quantum sonification, encapsulating most existing work. doi.org/10.1142/14025

Sonifying Quantum Superpositions by Correlating State Probability to Rhythmic Density 2023
Psychoacoustics-informed quantum sonification method, including musical and pedagogical considerations. Presented at ISQCMC 2023. doi.org/10.5281/zenodo.10206655

AWARDS / HONORS / SCHOLARSHIPS

Luddy School of Informatics, Computing, and Engineering Dean's List	2023 - 2025
Luddy Dean's Advisory Council Senior Scholarship (Indiana University Bloomington)	2024 - 2025
Bernhard Heiden Scholarship (Indiana University Bloomington)	2024 - 2025
Sarah Brown Sherman Scholarship (Indiana University Bloomington)	2023 - 2025
National Merit Scholarship (Indiana University Bloomington)	2021 - 2025
Martin Eliot Grey Scholarship (Indiana University Bloomington)	2021 - 2025
Music Faculty Award Scholarship (Indiana University Bloomington)	2021 - 2025
Sigma Alpha Iota Inter-American Music Awards Winner	2025
Premiere Young Artist Award Scholarship (Indiana University Bloomington)	2021 - 2023
RIOCC Organ Composition Competition Young Artist Division 1st Prize	2021